

posite of terse — long, perhaps, for the technical content contained in it. But I believe this is the key to reaching a wide audience not inclined (or not yet prepared) for reading terse technical mathematics.

Audience

I am aiming this book at anyone who wants to delve into some contemporary mathematics that is not done in school, and is very different from the kind of math done in school. This is regardless of your previous math achievement and your future math goals. I will not assume any knowledge or any recollection of school math, and I will gradually build up the level of formality in case it is the symbols and notation that have previously put you off. Here are some different types of reader I imagine:

- Adults who regret being put off math in the past and think, deep down, that they should be able to understand math if it's presented differently.
- Adults who always liked math and miss it, and feel like having some further mathematical stimulation.
- Anyone who wishes to learn some contemporary mathematics not covered in the standard curriculum, though I hope it might be one day.
- Math teachers who want to extend their knowledge and/or get ideas for how to teach abstract math without much (or any) prerequisite.
- School students who want extra math to stretch themselves, and/or an introduction into higher level math different from what students are usually stretched with at school.
- School students who are unhappy with their math classes and who might benefit from seeing a more profound, less routine type of math.
- Non-mathematicians who want to learn category theory but find the existing textbooks beyond them. Judging from correspondence since *How to Bake π* this might include programmers, engineers, business people, psychologists, linguists, writers, artists and more.
- Undergraduate mathematicians who have heard that category theory is important but are not sure why such abstraction is necessary, or are not sure how to approach it.
- Those with a math degree who would still like to have a gentle companion book to the existing texts, that contains more of the spirit of category theory alongside the technicalities.
- Home-schools and summer camps.

How to Bake π is not exactly a prerequisite but having read it will almost certainly help.

This material is developed from my teaching art students at the School of the Art Institute of Chicago. Most of the students had bad experiences of school math and many of them either can't remember any of it or have deliberately forgotten all of it as they found it so traumatic. This book seeks to be different from all of those types of experiences. It might seem long in terms of pages, but I hope you will quickly find that you can get through the pages much faster than you can for a standard math textbook. If the content here were written in a standard way it might only take 100 pages. I didn't want to make it shorter by explaining things less, so I have made it longer by explaining things more fully. I will gradually introduce formal mathematical language, and have included a glossary at the end for quick reference. I occasionally include the names of related concepts that are beyond the scope of this book, not because I think you need to know them, but in case you are interested and would like to look them up.

One obstacle to non-mathematicians trying to learn category theory is that the examples used are often taken from other parts of pure mathematics. In this book I will be sure to use examples that do not do that, including examples from daily experience such as family relationships, train journeys, freezing and thawing food, and more hard-hitting subjects such as racism and privilege. I have found that this helps non-mathematicians connect with abstract mathematics in ways that mathematical examples do not. Where I do include mathematical examples I will either introduce them fully as new concepts, or point out where they are not essential for continuing but are included for interest for those readers who have heard of them.

In particular, if you think you're bad at mental arithmetic, terrible at algebra, can't solve equations, and shudder at the thought of sketching graphs, that need not be an obstacle for you to read this book. I am not saying that you will find the book easy: abstraction is a way of thinking that takes some building of ability. We will build up through Part One of the book, and definitely take off in Part Two. It should be intellectually stretching, otherwise we wouldn't have achieved anything. But your previous experiences with math need not bar your way in here as they might previously have seemed to do. Most of all, aside from the technicalities of category theory I want to convey the joy I feel in the subject: in learning it, researching it, using it, applying it, thinking about it. More than technical prowess or a huge litany of theorems, I want to share the joy of abstraction.

PART ONE

BUILDING UP TO CATEGORIES

